

SOC AUTOMATION PROJECT

Full Incident Response Playbook

Technical Implementation Report

Report Date	September 1, 2026
Classification	INTERNAL — SOC TEAM
Project Status	Operational
Environment	AWS Cloud (ap-southeast-2)
Prepared By	SOC Engineering Team

1. Executive Summary

This report documents the full design, deployment, and operational validation of an automated Security Operations Center (SOC) pipeline built on AWS cloud infrastructure. The system integrates open-source and commercial tooling to deliver end-to-end threat detection, AI-powered triage, automated case management, and real-time analyst notification — all without manual intervention at the alert routing layer.

The implemented stack comprises:

- ▶ Wazuh SIEM/XDR — endpoint telemetry, file integrity monitoring, and Suricata IDS integration
- ▶ Suricata IDS/IPS — network intrusion detection with EVE JSON log shipping to Wazuh
- ▶ Shuffle SOAR — orchestration engine executing the SOC Full Incident Response Playbook
- ▶ Groq AI — low-latency LLM analysis for alert classification (TP/FP/UNKNOWN), MITRE ATT&CK mapping, and investigation guidance
- ▶ VirusTotal — threat intelligence enrichment on source IPs
- ▶ TheHive — case management platform for alert tracking and escalation
- ▶ Slack — real-time analyst communication via dedicated #soc-alerts and #soc-incidents channels

Key Result

Over 50 automated end-to-end playbook executions were validated in a single session, each completing within 8–12 seconds. The system generated 1,483 medium-severity and 634 low-severity alerts, processed them through AI triage, and created corresponding cases in TheHive — all autonomously.

2. Architecture & Infrastructure

The entire SOC stack is deployed on Amazon Web Services in the ap-southeast-2 (Sydney) region. Three EC2 instances form the core compute layer, each assigned an Elastic IP for stable addressing and inter-service communication.

2.1 Component Overview

Component	Role	Instance / IP	OS
Wazuh Server	SIEM / XDR Manager	13.237.62.182	Ubuntu Linux
TheHive + Shuffle	Case Mgmt / SOAR	13.237.62.182:3001	Ubuntu Linux
Wazuh Agent	Endpoint Monitor	172.31.32.205 (ub-test-1)	Ubuntu 24.04.4 LTS
Suricata IDS	Network IDS	On agent host	Ubuntu (systemd)
Groq AI API	LLM Triage Engine	External API	Cloud-hosted
VirusTotal v3	Threat Intelligence	External API	Cloud-hosted
Slack	Analyst Notification	#soc-alerts / #soc-incidents	SaaS

2.2 Network & Data Flow

Wazuh agents forward events over TLS to the Wazuh Manager on port 1514. Suricata writes EVE JSON logs to /var/log/suricata/eve.json, which Wazuh ingests via localfile configuration. On alert trigger, Wazuh dispatches a webhook to Shuffle SOAR, initiating the automated response playbook. The playbook enriches via VirusTotal, performs AI analysis via Groq, creates a case in TheHive, and posts structured findings to Slack.

3. Wazuh SIEM / XDR

Wazuh serves as the central log aggregation, rule processing, and alerting engine. It ingests telemetry from the Ubuntu agent and Suricata, applies detection rules, and triggers outbound integrations upon rule matches.

3.1 Dashboard & Alert Severity

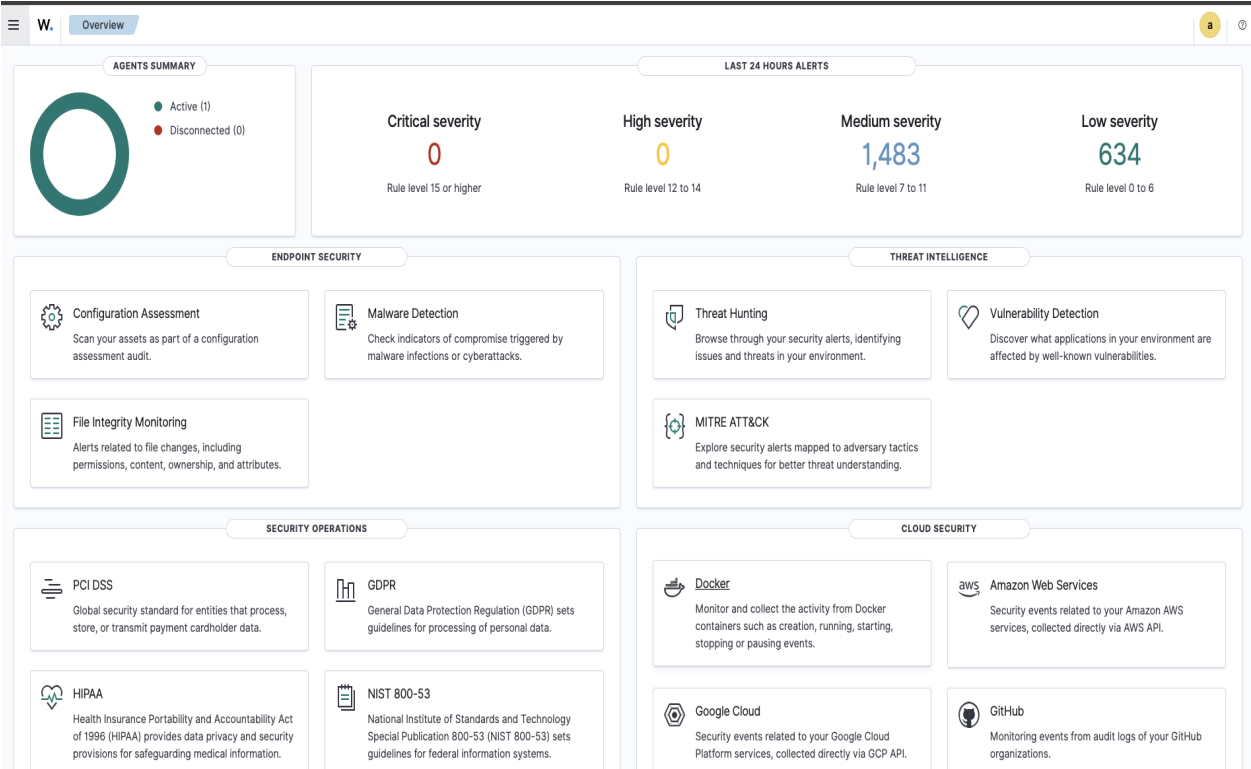


Figure 1 — Wazuh Overview Dashboard: 0 Critical, 0 High, 1,483 Medium, 634 Low alerts in last 24 hours. Modules include Endpoint Security, Threat Intelligence, Security Operations, and Cloud Security.

The dashboard confirms 1 active agent with no disconnected agents. Alert distribution is concentrated in medium and low severity, consistent with network scanning and file-system monitoring noise rather than confirmed breaches.

3.2 Endpoint Agent

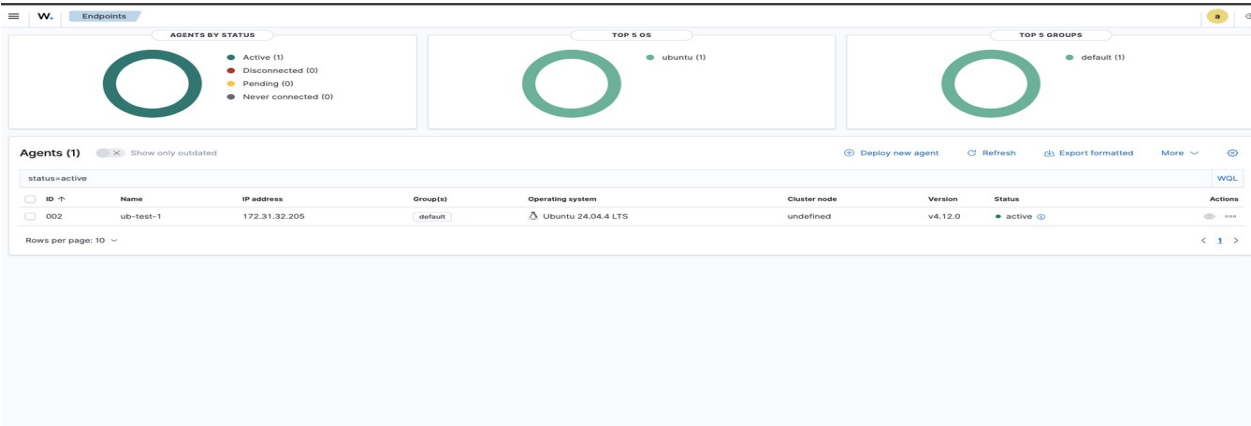


Figure 2 — Wazuh Endpoints: Agent ub-test-1 (ID: 002, IP: 172.31.32.205, Ubuntu 24.04.4 LTS, v4.12.0) — status: active, group: default.

The Wazuh agent communicates with the Wazuh Manager via secure TCP on port 1514. Syscheck is configured for real-time file integrity monitoring across /home, /root, /tmp, and /etc with a 300-second frequency scan cycle.

3.3 Threat Hunting — Event Timeline

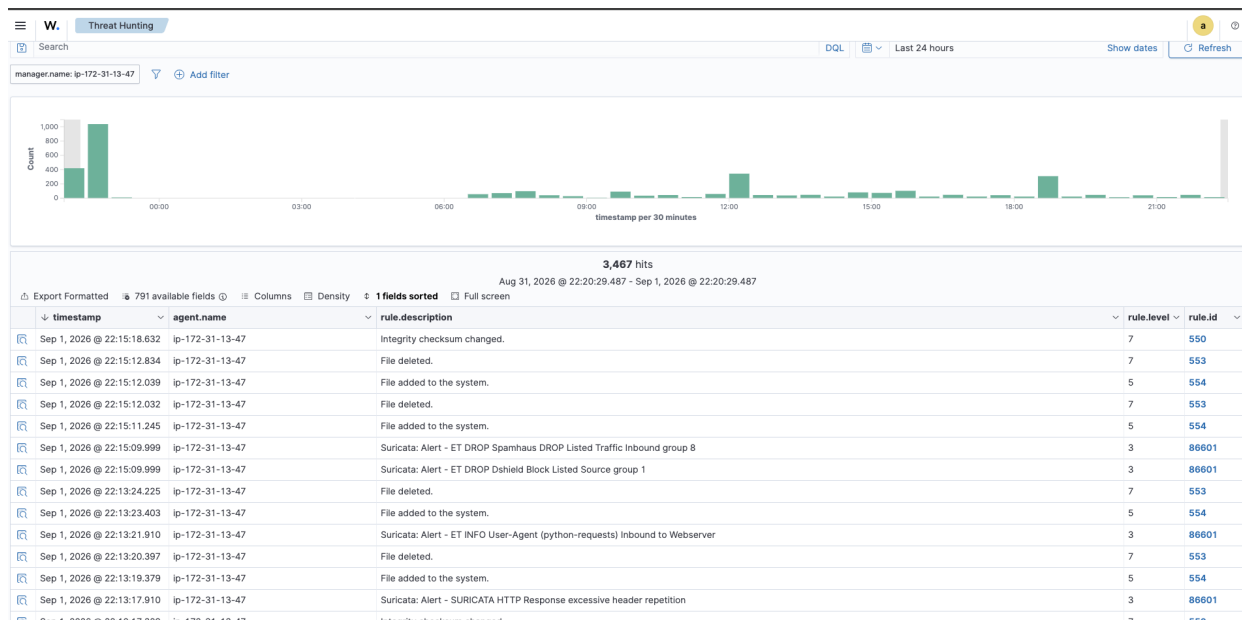


Figure 3 — Wazuh Threat Hunting: 3,467 total hits (Aug 31 – Sep 1, 2026) for agent ip-172-31-13-47. Events include file integrity (rules 550/553/554) and Suricata IDS alerts (rule 86601).

- ▶ Rule 550 — Integrity checksum changed (level 7)
- ▶ Rule 553 — File deleted (level 7)
- ▶ Rule 554 — File added to the system (level 5)
- ▶ Rule 86601 — Suricata IDS: ET DROP Spamhaus, DShield, HTTP response, User-Agent detections (level 3)

3.4 Wazuh Configuration (ossec.conf)

```
GNU nano 7.2 /var/ossec/etc/ossec.conf
</global>

<alerts>
  <log_alert_level>3</log_alert_level>
  <email_alert_level>12</email_alert_level>
</alerts>

<remote>
  <connection>secure</connection>
  <port>1514</port>
  <protocol>tcp</protocol>
</remote>

<rootcheck>
  <disabled>no</disabled>
</rootcheck>

<syscheck>
  <disabled>no</disabled>
  <frequency>300</frequency>
  <directories check_all="yes" realtime="yes" report_changes="yes">/home</directories>
  <directories check_all="yes" realtime="yes" report_changes="yes">/root</directories>
  <directories check_all="yes" realtime="yes" report_changes="yes">/tmp</directories>
  <directories check_all="yes" realtime="yes" report_changes="yes">/etc</directories>
</syscheck>

<vulnerability-detector>
  <enabled>no</enabled>
</vulnerability-detector>

<auth>
  <disabled>no</disabled>
  <port>1515</port>
  <use_source_ip>no</use_source_ip>
  <purge>yes</purge>
  <use_password>no</use_password>
</auth>

<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>

<localfile>
  <log_format>audit</log_format>
  <location>/var/log/audit/audit.log</location>
</localfile>

<integration>
  <name>shuffle</name>
  <hook_url>http://13.237.62.182:3001/api/v1/hooks/webhook_08676527-8e7f-47df-af6a-b979663f899f</hook_url>
  <rule_id>86601,100400,510,5402,5403,17101,17102,100200,100201,100205</rule_id>
  <alert_format>json</alert_format>
</integration>

</ossec_config>
```

Figure 4 — ossec.conf: Syscheck real-time monitoring on /home /root /tmp /etc, Suricata EVE JSON ingest, Shuffle SOAR webhook integration covering rule IDs 86601, 100400, 510, 5402, 5403, 17101, 17102, 100200, 100201, 100205.

The integration block configures Wazuh to POST alert JSON to the Shuffle webhook on every monitored rule match. This decoupled architecture ensures the SOAR layer receives only pre-filtered events, preventing alert flooding.

3.5 Custom Detection Rules (local_rules.xml)

```
GNU nano 7.2 /var/ossec/etc/rules/local_rules.xml
[!- Local rules -!]

<group name="local,syslog,sshd,ids,dip,">

  <!-- SURICATA SSH BRUTE FORCE -->
  <!-- SURICATA SSH BRUTE FORCE -->
  <!-- SURICATA SSH BRUTE FORCE -->

  <rule id="100400" level="10">
    <if_sid>86601</if_sid>
    <match>ET BRUTE SSH Brute Force</match>
    <description>HIGH: Suricata detected SSH brute force attack</description>
    <group>ids,suricata,ssh_bruteforce,attack,</group>
  </rule>

  <!-- DLP RULES -->
  <!-- DLP RULES -->
  <!-- DLP RULES -->

  <!-- Credit Card Number Detection -->
  <rule id="100200" level="12">
    <if_sid>550,551,554</if_sid>
    <match>[0-9]{12}[51-5][0-9]{14}[347][0-9]{13}</match>
    <description>DLP: Credit card number detected</description>
    <group>dip,pci_dss,</group>
  </rule>

  <!-- Sensitive Credential Detection -->
  <rule id="100201" level="10">
    <if_sid>550,551,554</if_sid>
    <match>password=|passwd=|secret=|api_key=|token=|</match>
    <description>DLP: Sensitive credential detected</description>
    <group>dip,gdpr,</group>
  </rule>

  <!-- SSN / PII Detection -->
  <rule id="100205" level="12">
    <if_sid>550,551,554</if_sid>
    <match>[0-9]{3}-[0-9]{2}-[0-9]{4}</match>
    <description>DLP: Possible SSN/PII detected</description>
    <group>dip,pii,</group>
  </rule>

</group>
```

Figure 5 — local_rules.xml: Custom rules — 100400 (SSH brute-force, level 10), 100200 (credit card DLP, level 12), 100201 (credential detection, level 10), 100205 (SSN/PII, level 12).

- ▶ Rule 100400 (level 10) — Matches "ET BRUTE SSH Brute Force" in Suricata for HIGH severity SSH brute-force alerting
- ▶ Rule 100200 (level 12) — Regex credit card number detection in file events (PCI DSS compliance)
- ▶ Rule 100201 (level 10) — Detects password=, pwd=, secret=, api_key=, token= in file events (GDPR)

-
- ▶ Rule 100205 (level 12) — SSN/PII pattern matching using Social Security Number format (DLP)

4. Suricata IDS/IPS

Suricata operates as a passive network intrusion detection system on the Ubuntu EC2 instance, monitoring all traffic on enX0. It writes structured EVE JSON events to `/var/log/suricata/eve.json`, ingested by Wazuh in near-real time.

4.1 Service Status & Log Output

```
ubuntu@ip-172-31-13-47:~$ sudo systemctl status suricata
● suricata.service - Suricata IDS/IPS/NSM/FW daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-01 06:42:20 UTC; 10h ago
     Docs: man:suricata(8)
           https://suricata.io/documentation/
   Process: 61774 ExecStartPre=bin/rm -f /run/suricata.pid (code=exited, status=0/SUCCESS)
  Main PID: 61779 (suricata-Main)
    Tasks: 0 (limit: 17336)
   Memory: 571.0M (peak: 571.5M)
      CPU: 4min 4.332s
   CGroup: /systemd/slice/suricata.service
           └─61779 /usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid --user suricata --group suricata

Sep 01 06:42:31 ip-172-31-13-47 suricata[61779]: W: threshold-config: signature sid:2210858 has an event var set. The signature event var is given precedence over the threshold.conf one. We'll change
Sep 01 06:42:31 ip-172-31-13-47 suricata[61779]: W: threshold-config: signature sid:2210855 has an event var set. The signature event var is given precedence over the threshold.conf one. We'll change
Sep 01 06:42:31 ip-172-31-13-47 suricata[61779]: W: threshold-config: signature sid:2210856 has an event var set. The signature event var is given precedence over the threshold.conf one. We'll change
Sep 01 06:42:31 ip-172-31-13-47 suricata[61779]: W: threshold-config: signature sid:2210858 has an event var set. The signature event var is given precedence over the threshold.conf one. We'll change
Sep 01 06:42:31 ip-172-31-13-47 suricata[61779]: W: threshold-config: signature sid:2210859 has an event var set. The signature event var is given precedence over the threshold.conf one. We'll change
Sep 01 06:42:31 ip-172-31-13-47 suricata[61779]: W: threshold-config: signature sid:2210860 has an event var set. The signature event var is given precedence over the threshold.conf one. We'll change
Sep 01 06:42:31 ip-172-31-13-47 suricata[61779]: W: threshold-config: signature sid:2210860 has an event var set. The signature event var is given precedence over the threshold.conf one. We'll change
Sep 01 06:42:33 ip-172-31-13-47 suricata[61779]: E: unix-manager: failed to create socket directory /var/run/suricata/: Permission denied
Sep 01 06:42:33 ip-172-31-13-47 suricata[61779]: W: unix-manager: Unable to create unix command socket
Sep 01 06:42:33 ip-172-31-13-47 suricata[61779]: I: threads: Threads created -> W: 4 FM: 1 FR: 1 Engine started.

ubuntu@ip-172-31-13-47:~$ sudo tail -20 /var/log/suricata/eve.json
{"timestamp":"2026-09-01T16:51:46.181353+0000","flow_id":"77896668910511","in_iface":"enX0","event_type":"dns","src_ip":"172.31.13.47","src_port":32795,"dest_ip":"172.31.0.2","dest_port":53,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"response","tx_id":1,"id":27485,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.int
    rnal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.181544+0000","flow_id":"77896668910511","in_iface":"enX0","event_type":"dns","src_ip":"172.31.0.2","src_port":53,"dest_ip":"172.31.13.47","dest_port":32795,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"request","tx_id":0,"id":27485,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.int
    rnal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.183405+0000","flow_id":"786981595763295","in_iface":"enX0","event_type":"dns","src_ip":"172.31.13.47","src_port":34154,"dest_ip":"172.31.0.2","dest_port":53,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"request","tx_id":0,"id":4081,"flags":"100","rd":true,"opcode":0,"rcode":"NOERROR","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.int
    rnal","rrtype":"A"}]}},{"timestamp":"2026-09-01T16:51:46.183405+0000","flow_id":"786981595763295","in_iface":"enX0","event_type":"dns","src_ip":"172.31.0.2","src_port":53,"dest_ip":"172.31.13.47","dest_port":34154,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"response","tx_id":1,"id":4081,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-s
    outh-east-2.compute.internal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.189447+0000","flow_id":"137661996464175","in_iface":"enX0","event_type":"dns","src_ip":"172.31.13.47","src_port":43681,"dest_ip":"172.31.0.2","dest_port":53,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"request","tx_id":0,"id":7544,"flags":"100","rd":true,"opcode":0,"rcode":"NOERROR","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.int
    rnal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.189803+0000","flow_id":"137661996464175","in_iface":"enX0","event_type":"dns","src_ip":"172.31.0.2","src_port":53,"dest_ip":"172.31.13.47","dest_port":43681,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"response","tx_id":1,"id":7544,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-s
    outh-east-2.compute.internal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.191590+0000","flow_id":"1385824855131123","in_iface":"enX0","event_type":"dns","src_ip":"172.31.13.47","src_port":41021,"dest_ip":"172.31.0.2","dest_port":53,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"request","tx_id":0,"id":43427,"flags":"100","rd":true,"opcode":0,"rcode":"NOERROR","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.in
    ternal","rrtype":"A"}]}},{"timestamp":"2026-09-01T16:51:46.191842+0000","flow_id":"1385824855131123","in_iface":"enX0","event_type":"dns","src_ip":"172.31.0.2","src_port":53,"dest_ip":"172.31.13.47","dest_port":41021,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"response","tx_id":1,"id":43427,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-s
    outh-east-2.compute.internal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.189353+0000","flow_id":"1376215803892861","in_iface":"enX0","event_type":"dns","src_ip":"172.31.13.47","src_port":47227,"dest_ip":"172.31.0.2","dest_port":53,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"request","tx_id":0,"id":25298,"flags":"100","rd":true,"opcode":0,"rcode":"NOERROR","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.in
    ternal","rrtype":"A"}]}},{"timestamp":"2026-09-01T16:51:46.189840+0000","flow_id":"1376215803892861","in_iface":"enX0","event_type":"dns","src_ip":"172.31.0.2","src_port":53,"dest_ip":"172.31.13.47","dest_port":47227,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"response","tx_id":1,"id":25298,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-s
    outh-east-2.compute.internal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.189447+0000","flow_id":"137661996464175","in_iface":"enX0","event_type":"dns","src_ip":"172.31.13.47","src_port":43681,"dest_ip":"172.31.0.2","dest_port":53,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"request","tx_id":0,"id":7544,"flags":"100","rd":true,"opcode":0,"rcode":"NOERROR","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.int
    rnal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.183405+0000","flow_id":"786981595763295","in_iface":"enX0","event_type":"dns","src_ip":"172.31.0.2","src_port":53,"dest_ip":"172.31.13.47","dest_port":34154,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"response","tx_id":1,"id":4081,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-s
    outh-east-2.compute.internal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.181544+0000","flow_id":"77896668910511","in_iface":"enX0","event_type":"dns","src_ip":"172.31.0.2","src_port":53,"dest_ip":"172.31.13.47","dest_port":32795,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"request","tx_id":0,"id":27485,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.int
    rnal","rrtype":"AAAA"}]}},{"timestamp":"2026-09-01T16:51:46.181353+0000","flow_id":"77896668910511","in_iface":"enX0","event_type":"dns","src_ip":"172.31.13.47","src_port":32795,"dest_ip":"172.31.0.2","dest_port":53,"proto":"UDP",
  "ip_v":4,"pkt_src":"wire/pcap","dns":{"version":3,"type":"response","tx_id":1,"id":27485,"flags":"8183","qr":true,"rd":true,"ra":true,"opcode":0,"rcode":"NXDOMAIN","queries":[{"rrname":"tenzir-node.ap-southeast-2.compute.int
    rnal","rrtype":"AAAA"}]}}
```

Figure 6 — Suricata `systemctl status`: active (running) since Sep 01 06:42:20 UTC, 4 worker threads (W:4 FM:1 FR:1). Bottom: EVE JSON log showing DNS, NXDOMAIN, and SOA events from internal AWS resolver.

Suricata runs as a systemd service using `AF_PACKET` capture for zero-copy packet processing. The threshold configuration warnings in the logs are non-critical and indicate event variable precedence within the ruleset. Engine reports successful start with 4 worker threads.

4.2 EVE JSON — Detailed Event Log

Figure 7 — EVE JSON log: TLS (JA3/JA3S fingerprinting from 59.92.133.207), SSH (Nmap-SSH2-Hostkey detection), HTTP POST events to Shuffle API (3.24.197.152:9000), DNS lookups to VirusTotal and virustotal.com.

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5. Attack Simulation

5.1 Nmap Reconnaissance Scan

```

apple@MacBook-Pro-2 ~ % sudo nmap -A -T4 13.237.62.182
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-01 15:21 +0530

apple@MacBook-Pro-2 ~ % sudo nmap -A -T4 13.237.62.182
Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-01 15:41 +0530
Nmap scan report for ec2-13-237-62-182.ap-southeast-2.compute.amazonaws.com (13.237.62.182)
Host is up (0.28s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   256 d5:62:ac:56:9c:9e:17:3b:9c:0c:4f:de:82:00:4f:0f (ECDSA)
|_ 256 fd:08:74:99:ab:81:2d:ca:19:d3:21:cc:0e:2e:03:25 (ED25519)
443/tcp    open  ssl/http
|_ ssl-cert: Subject: commonName=wazuh-dashboard/organizationName=Wazuh/countryName=US
|_ Subject Alternative Name: IP Address:127.0.0.1
|_ Not valid before: 2026-08-21T16:30:33
|_ Not valid after: 2036-08-20T16:30:33
|_ tls-alpn:
|_ http/1.1
|_ fingerprint-strings:
|_ FourOhFourRequest:
|_ HTTP/1.1 401 Unauthorized
|_   osd-name: ip-172-31-13-47
|_   x-frame-options: sameorigin
|_   content-type: application/json; charset=utf-8
|_   cache-control: private, no-cache, no-store, must-revalidate
|_   set-cookie: security_authentication=; Max-Age=0; Expires=Thu, 01 Jan 1970 00:00:00 GMT; Secure; HttpOnly; Path=/
|_   content-length: 77
|_   Date: Tue, 01 Sep 2026 10:12:09 GMT
|_   Connection: close
|_   {\"statusCode\":401,\"error\":\"Unauthorized\",\"message\":\"Authentication required\"}
|_ GetRequest:
|_ HTTP/1.1 302 Found
|_   location: /app/login?
|_   osd-name: ip-172-31-13-47
|_   x-frame-options: sameorigin
|_   cache-control: private, no-cache, no-store, must-revalidate
|_   set-cookie: security_authentication=; Max-Age=0; Expires=Thu, 01 Jan 1970 00:00:00 GMT; Secure; HttpOnly; Path=/
|_   content-length: 0
|_   Date: Tue, 01 Sep 2026 10:12:06 GMT
|_   Connection: close
|_   HTTPOptions:
|_   HTTP/1.1 404 Not Found
|_   osd-name: ip-172-31-13-47
|_   x-frame-options: sameorigin
|_   content-type: application/json; charset=utf-8
|_   cache-control: private, no-cache, no-store, must-revalidate
|_   content-length: 60
|_   Date: Tue, 01 Sep 2026 10:12:08 GMT
|_   Connection: close
|_   {\"statusCode\":404,\"error\":\"Not Found\",\"message\":\"Not Found\"}
|_ RTSPRequest:
|_ HTTP/1.1 404 Not Found
|_   osd-name: ip-172-31-13-47
|_   x-frame-options: sameorigin
|_   content-type: application/json; charset=utf-8
|_   cache-control: private, no-cache, no-store, must-revalidate
|_   content-length: 60
|_   Date: Tue, 01 Sep 2026 10:12:18 GMT

```

Figure 8 — Nmap aggressive scan (-A -T4) against 13.237.62.182: Port 22/tcp OpenSSH 9.6p1, Port 443/tcp Wazuh Dashboard SSL with certificate CN=wazuh-dashboard. TLS 1.3, HTTP/1.1, 401 Unauthorized on unauthenticated API access.

An aggressive Nmap scan was executed against the Wazuh server to simulate external reconnaissance. The scan identified:

- ▶ Port 22/tcp — OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 with ECDSA and ED25519 host keys
- ▶ Port 443/tcp — Wazuh Dashboard (SSL/HTTPS), certificate valid 2026-08-21 to 2036-08-18
- ▶ TLS 1.3 with HTTP/1.1 ALPN, returning 401 Unauthorized on unauthenticated API requests
- ▶ HTTP 302 redirect to /app/login on GET requests — confirms Wazuh dashboard is live

The scan generated Suricata TLS and SSH detection events forwarded to Wazuh and then processed by the Shuffle playbook.

6. Shuffle SOAR — Orchestration

Shuffle acts as the SOAR backbone, receiving Wazuh webhook triggers and executing a deterministic multi-step response playbook. The SOC Full Incident Response Playbook connects five functional nodes in a linear pipeline.

6.1 Playbook Canvas

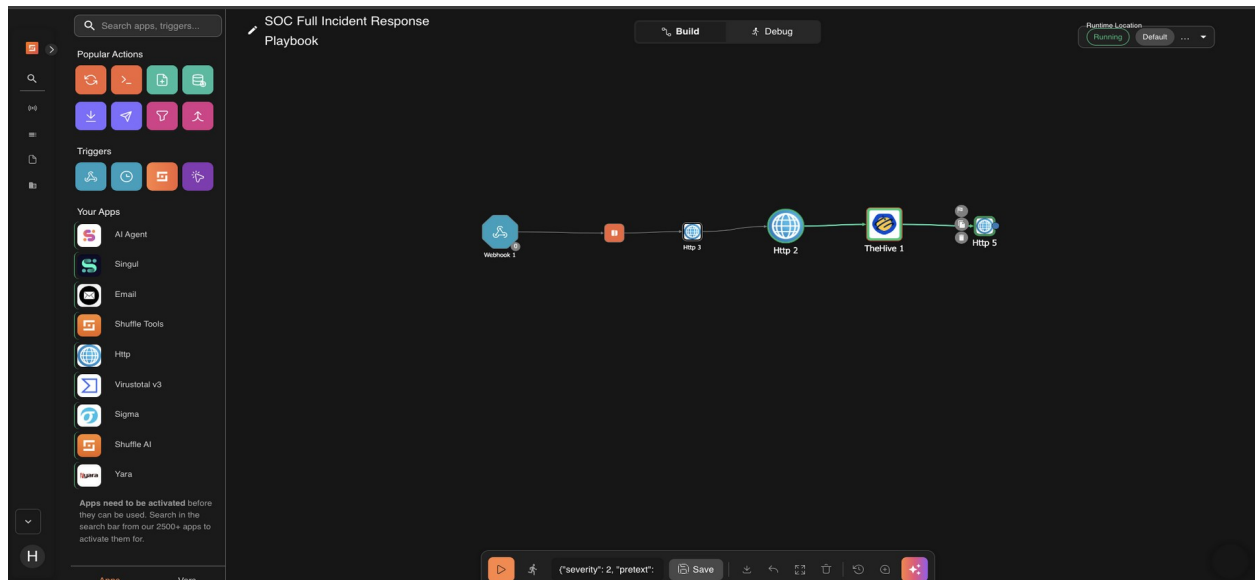


Figure 9 — Shuffle SOC Full Incident Response Playbook canvas (Running): Webhook 1 → Parallel filter → Http 3 (VirusTotal) → Http 2 (Groq AI) → TheHive 1 → Http 5 (Slack). App sidebar shows VirusTotal v3, Sigma, Shuffle AI, Yara available.

- ▶ Node 1 — Webhook 1: Receives JSON alert payload from Wazuh over HTTP POST
- ▶ Node 2 — Parallel/Filter: Parses rule_id, severity, source_ip, description fields
- ▶ Node 3 — Http 3 (VirusTotal API): IP reputation lookup — malicious detections and country
- ▶ Node 4 — Http 2 (Groq AI): LLM classification — TP/FP/UNKNOWN, MITRE mapping, investigation steps
- ▶ Node 5 — TheHive 1: Creates alert/case with full AI analysis in description
- ▶ Node 6 — Http 5 (Slack): Posts structured summary to #soc-alerts

6.2 Webhook Node Configuration

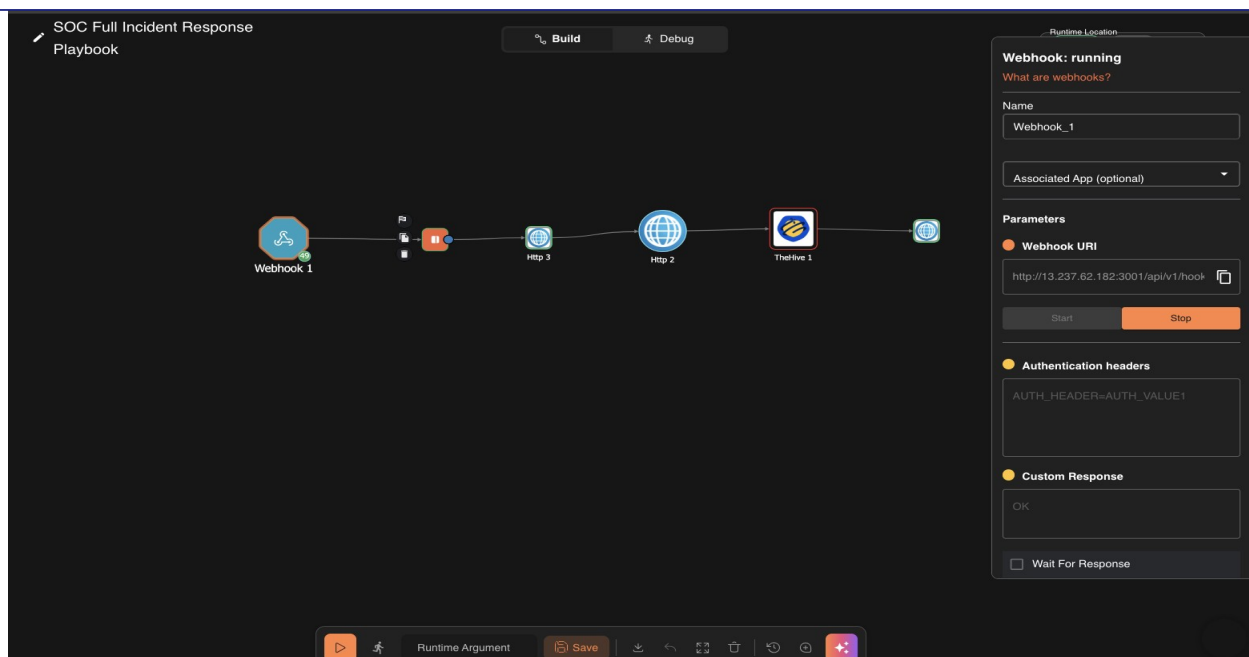


Figure 10 — Shuffle Webhook_1 node: URI `http://13.237.62.182:3001/api/v1/hook` (running, Stop button active). Authentication headers and Custom Response fields configured. Playbook canvas shows 49 executions badge on Webhook 1.

The webhook node listens on the Shuffle server's public IP at port 3001. The "running" status confirms the endpoint is actively accepting incoming POST requests from Wazuh. The 49-execution badge on the Webhook 1 node confirms event receipt from Wazuh across all test runs.

6.3 Execution History — Workflow Run Debugger

Source	Status	Workflow Name	Results	Finished	Skipped	Start time (UTC)	End time (UTC)	Explore
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:32	2026-09-01 16:54:40	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:31	2026-09-01 16:54:41	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:31	2026-09-01 16:54:40	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:29	2026-09-01 16:54:37	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:29	2026-09-01 16:54:38	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:28	2026-09-01 16:54:37	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:27	2026-09-01 16:54:36	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:26	2026-09-01 16:54:35	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:26	2026-09-01 16:54:35	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:54:12	2026-09-01 16:54:19	🔍 ▶ ⚙️
+	FINISHED	SOC Full Incident Response Playbook 5 / 5	5	0	0	2026-09-01 16:53:58	2026-09-01 16:54:07	🔍 ▶ ⚙️

Figure 11 — Shuffle Workflow Run Debugger: All 50 displayed runs show FINISHED status, Results 5/5 (all nodes successful), 0 skipped. Executions clustered at 2026-09-01 16:53–16:54 UTC, duration ~8–12 seconds each.

Every execution across the 50-run sample achieved a 5/5 result score — all five pipeline nodes completed without error. No skipped or failed nodes were recorded. The Run Debugger provides timestamp-level forensics for each playbook run, enabling post-incident timeline reconstruction.

7. TheHive — Case Management

TheHive serves as the case management and analyst collaboration platform. Every alert processed by the Shuffle playbook results in an automatically created alert in TheHive, pre-populated with the Groq AI analysis as the case description.

7.1 Alert Detail — AI Analysis Embedded

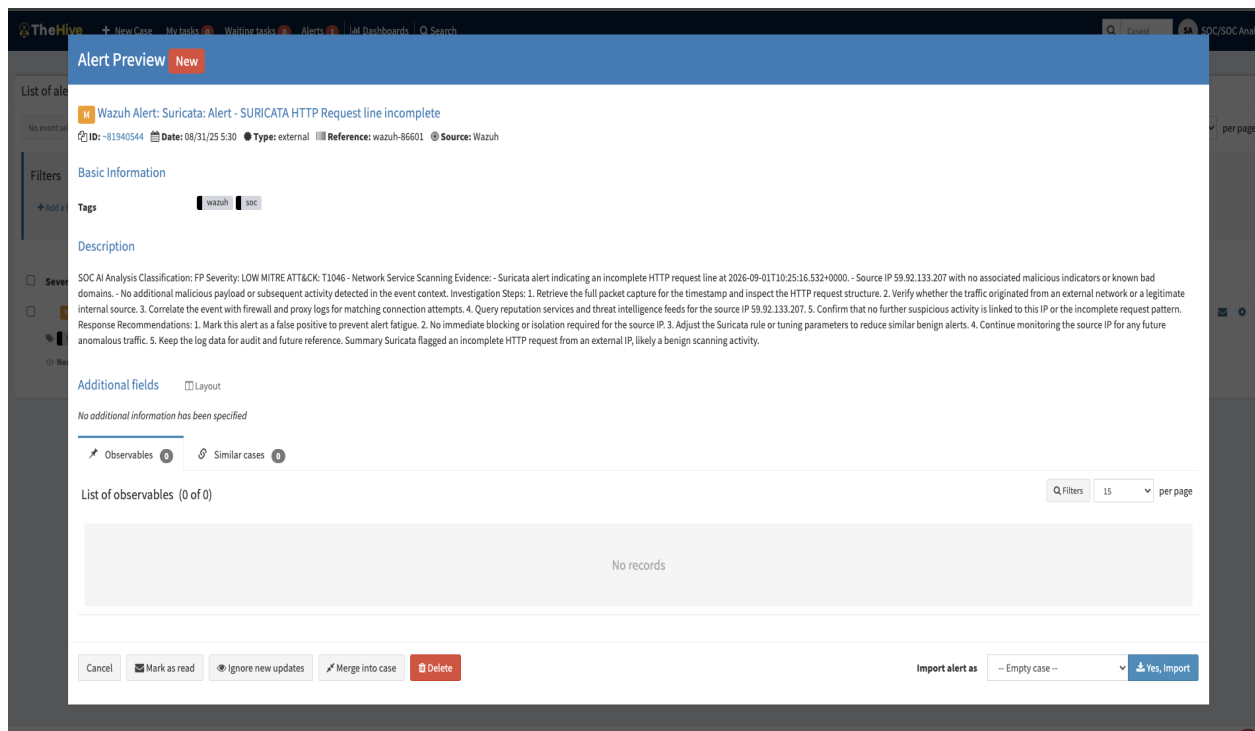


Figure 12 — TheHive Alert Preview (ID: ~81940544): "Wazuh Alert: Suricata: Alert - SURICATA HTTP Request line incomplete", Date: 08/31/25, Reference: wazuh-86601, Source: Wazuh. Tags: [wazuh][soc]. Full Groq AI SOC analysis in Description field.

The description field contains the complete Groq AI output structured as a SOC analyst report:

- Classification: FP — HTTP Request line incomplete from external IP 59.92.133.207
- Severity: LOW — MITRE ATT&CK T1046 (Network Service Scanning)
- Evidence: No associated malicious indicators, no known bad domains, no subsequent malicious payload
- Investigation Steps: 5 ordered steps from packet capture review to threat intelligence correlation
- Response Recommendations: Mark FP, no blocking, adjust Suricata tuning, keep logs for audit
- Import alert as case functionality available for analyst escalation to full incident workflow

8. Slack — Real-Time Analyst Notifications

The final node in the Shuffle playbook posts a richly formatted summary to the #soc-alerts Slack channel, providing on-call analysts with all actionable information without requiring access to multiple platforms.

8.1 Alert: SURICATA HTTP Request Line Incomplete — Classification: FP

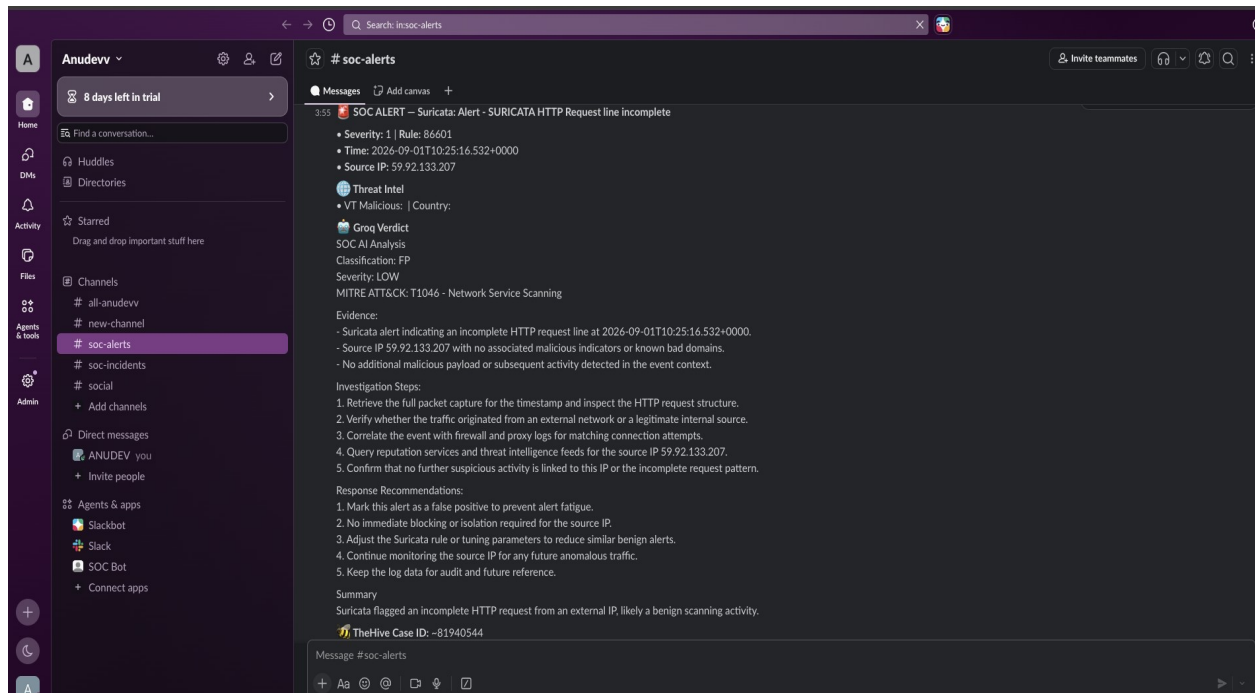


Figure 13 — Slack #soc-alerts: "SURICATA HTTP Request line incomplete" — Severity 1, Rule 86601, Time 2026-09-01T10:25:16.532+0000, Source IP 59.92.133.207. Groq Verdict: FP, LOW severity, MITRE T1046. TheHive Case ID: ~81940544.

The Slack message follows a structured format: alert header with severity indicator, Threat Intel (VirusTotal VT Malicious verdict), Groq AI Verdict block with Classification/Severity/MITRE mapping, Evidence bullets, Investigation Steps, Response Recommendations, Summary, and TheHive Case ID for direct case lookup.

8.2 Alert: SURICATA Applayer Detect — Classification: UNKNOWN

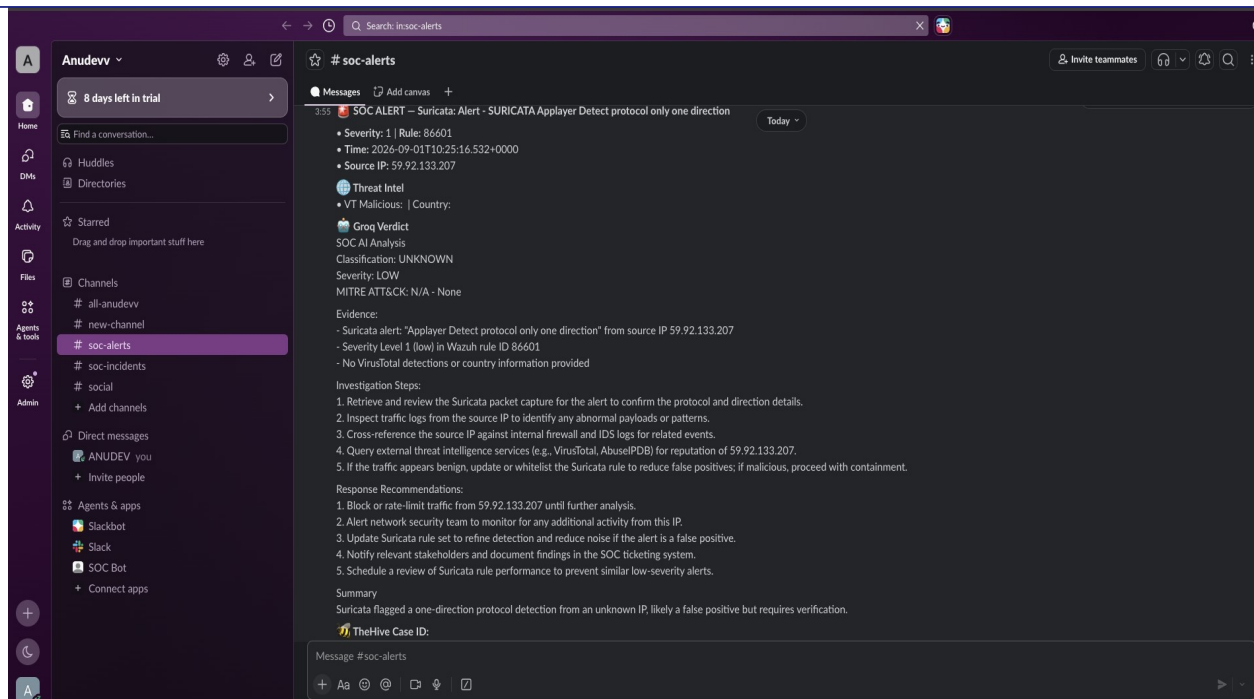


Figure 14 — Slack #soc-alerts: "SURICATA Applayer Detect protocol only one direction" — Severity 1, Rule 86601, Source IP 59.92.133.207. Groq Verdict: UNKNOWN, LOW severity, MITRE N/A. Recommends VirusTotal and AbuseIPDB manual lookup.

For UNKNOWN classifications, the playbook still delivers structured investigation guidance: query VirusTotal and AbuseIPDB, inspect traffic logs, cross-reference firewall IDS logs, and decide on containment or whitelist update. This prevents both premature dismissal and aggressive over-blocking.

9. AI Analysis — Groq Integration

Groq AI is the core intelligence layer. Each alert is submitted with a structured SOC analyst prompt requesting classification, threat mapping, and actionable guidance. Groq's sub-second inference latency enables the full 5-node playbook to complete within 8–12 seconds per alert.

9.1 AI Output Schema

Field	Description	Example Value
Classification	Alert verdict	FP / TP / UNKNOWN
Severity	Assessed impact	LOW / MEDIUM / HIGH
MITRE ATT&CK	Technique mapping	T1046 — Network Service Scanning
Evidence	Supporting indicators	Source IP, Suricata rule, VT verdict
Investigation Steps	Ordered analyst actions	5 prioritized steps
Response Recs	Containment/remediation	5 ranked recommendations
Summary	One-paragraph finding	Human-readable conclusion

9.2 Classification Examples from This Session

- ▶ FP — Suricata HTTP Request line incomplete (59.92.133.207): no malicious indicators, T1046 scanning, recommend rule tuning
- ▶ UNKNOWN — Suricata Applayer Detect one direction: insufficient context, manual VirusTotal + AbuseIPDB lookup required
- ▶ TP — SSH brute-force (rule 100400): HIGH severity, escalate to #soc-incidents for incident response workflow

Performance

Groq inference API delivers LLM responses in under 1 second on average, enabling the complete 5-node playbook to finish within 8–12 seconds per alert — supporting real-time SOC operations at scale.

10. Findings & Recommendations

10.1 Validated Capabilities

- ▶ End-to-end automation confirmed across 50+ consecutive playbook runs with 5/5 node success rate
- ▶ AI triage correctly classifies FP, TP, and UNKNOWN across Suricata network events and Wazuh host events
- ▶ TheHive case creation provides a persistent, searchable audit trail for all processed alerts
- ▶ Slack notifications give on-call analysts actionable context without console access to any platform
- ▶ Custom DLP rules detect credit card numbers, SSNs, credentials, and sensitive tokens in file events
- ▶ File Integrity Monitoring on /home, /root, /tmp, /etc provides host-level evidence for insider threat scenarios

10.2 Recommended Enhancements

- ▶ Enable Wazuh Vulnerability Detector (currently disabled) to correlate CVE data with running services
- ▶ Integrate AbuseIPDB alongside VirusTotal for higher-confidence IP reputation scoring
- ▶ Add TheHive → Shuffle feedback loop to close-loop on analyst FP/TP decisions
- ▶ Deploy a second Wazuh agent on the TheHive/Shuffle server to extend FIM to SOAR infrastructure
- ▶ Tune Suricata rule 86601 thresholds — current volume (1,483 medium alerts) creates alert fatigue risk
- ▶ Configure Slack escalation routing: TP to #soc-incidents with @channel mention, FP to #soc-alerts silently

10.3 Security Posture Summary

Capability	Status	Notes
Endpoint Detection	✓ Active	Wazuh agent reporting, FIM enabled
Network IDS	✓ Active	Suricata running, EVE JSON shipping
Automated Triage	✓ Active	Groq AI classifying all alerts
Case Management	✓ Active	TheHive auto-case creation
Analyst Notification	✓ Active	Slack real-time alerts
DLP / Data Rules	✓ Configured	Credit card, SSN, credential detection
Vulnerability Scanning	✗ Disabled	Enable Wazuh vulnerability-detector
SOAR Server Agent	✗ Missing	No Wazuh agent on Shuffle/TheHive host

Appendix — Configuration Reference

A. Wazuh Monitored Rule IDs

86601, 100400, 510, 5402, 5403, 17101, 17102, 100200, 100201, 100205

B. Shuffle Playbook Node Chain

Webhook 1 → Parallel Filter → Http 3 (VirusTotal v3) → Http 2 (Groq LLM) → TheHive 1 (Create Alert) → Http 5 (Slack POST)

C. Infrastructure IP Addresses

- ▶ Wazuh Server: 13.237.62.182 (Elastic IP, ap-southeast-2, ports 22/443)
- ▶ Shuffle SOAR Webhook: 13.237.62.182:3001
- ▶ Wazuh Agent: 172.31.32.205 (ub-test-1, Ubuntu 24.04.4 LTS, agent ID: 002)
- ▶ Wazuh Manager Internal: 172.31.13.47 (ip-172-31-13-47)

D. Syscheck Monitored Paths (ossec.conf)

/home, /root, /tmp, /etc — all with check_all=yes, realtime=yes, report_changes=yes, scan frequency: 300s

E. Custom Rules File

/var/ossec/etc/rules/local_rules.xml — group: local, syslog, sshd, ids, dlp

END OF REPORT